```
char L='a',  U='A';
```

b B

c C

```
for( r=1; r<=4; r++ )
{
for( c=1;c<=4; c++ )
{
if(r%2==0) p(L);
else  p(U);
}
if(r%2==0)L++; else  U++;
}
```

r

1

2

```
AAAA
aaaa
BBBB
bbbb
```

TC

File Edit Run Compile Project Options Debug Break/watch

Line 14 Col 39 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr, nc,r,c; char L='a', U='A';
clrscr();
printf("Enter no of rows and columns "); scanf("%d %d",&nr, &nc);
for(r=1;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
if(r%2==0)printf("%2c",L);  else  printf("%2c",U);
}
printf("\n"); if(r%2==0)L++; else U++;_
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC

```
Enter no of rows and columns 10 20
 A A A A A A A A A A A A A A A A A A A A
 a a a a a a a a a a a a a a a a a a a a
 B B B B B B B B B B B B B B B B B B B B
 b b b b b b b b b b b b b b b b b b b b
 C C C C C C C C C C C C C C C C C C C C
 c c c c c c c c c c c c c c c c c c c c
 D D D D D D D D D D D D D D D D D D D D
 d d d d d d d d d d d d d d d d d d d d
 E E E E E E E E E E E E E E E E E E E E
 e e e e e e e e e e e e e e e e e e e e
_
```

Activate Windows
Go to PC settings to activate Windows.

```
      Line 18    Col 1   Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr, nc,r,c;
clrscr();
printf("Enter no of rows and columns "); scanf("%d %d",&nr, &nc);
for(r=1;r<=nr;r++)
{
char L='a', U='A';
for(c=1;c<=nc;c++)
{
if(c%2==0)printf("%2c",L++);  else  printf("%2c",U++);
}
printf("\n");
}
getch();
}
```

```
 A a B b C c D d E e F f G g H h I i J j
 A a B b C c D d E e F f G g H h I i J j
 A a B b C c D d E e F f G g H h I i J j
 A a B b C c D d E e F f G g H h I i J j
 A a B b C c D d E e F f G g H h I i J j
 A a B b C c D d E e F f G g H h I i J j
 A a B b C c D d E e F f G g H h I i J j
 A a B b C c D d E e F f G g H h I i J j
 A a B b C c D d E e F f G g H h I i J j
 A a B b C c D d E e F f G g H h I i J j
```

```
char L='a',  U='A';
```

b B

c C

```
for( r=1; r<=4; r++ )
{   char L='a',  U='A';
for( c=1;c<=4; c++ )
{
if(c%2==0) p(L ++ );
else  p(U ++  );
}

}
```

b c

1 1

2 2

```
AaBbCc
AaBbCc
AaBbCc
```

TC

Line 18 Col 55 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr, nc,r,c;
clrscr();
printf("Enter no of rows and columns "); scanf("%d %d",&nr, &nc);
for(r=1;r<=nr;r++)
{
char ch=64+nc;
for(c=1;c<=nc;c++)
{
if(r%2==0)printf("%2c",ch--);  else  printf("%2c",64+c);
}
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC

```
Enter no of rows and columns 10 26
 A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
 Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
 A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
 Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
 A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
 Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
 A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
 Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
 A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
 Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
```

Activate Windows
Go to PC settings to activate Windows.

```
for( r=1; r<=4; r++ )
{
   ch=64+nc;
for( c=1;c<=4; c++ )
{
if(r%2==0) p(ch--);
else p(c+64);
```

nr	nc	ch
4	4	64+4=68D
		67C
		66B
		65A

```
ABCD
DCBA
ABCD
DCBA
```

TC

File Edit Run Compile Project Options Debug Break/watch

Line 12 Col 49 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&n);
for(r=1;r<=n;r++)
{
for(c=1;c<=n;c++)
{
if(c<=n-r)printf("%3d",1);  else  printf("%3d",r);
}
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC

```
Enter no of rows 4
  1  1  1  1
  1  1  2  2
  1  3  3  3
  4  4  4  4
```

Activate Windows
Go to PC settings to activate Windows.

```
Enter no of rows 10
  1  1  1  1  1  1  1  1  1  1
  1  1  1  1  1  1  1  1  2  2
  1  1  1  1  1  1  1  3  3  3
  1  1  1  1  1  1  4  4  4  4
  1  1  1  1  1  5  5  5  5  5
  1  1  1  1  6  6  6  6  6  6
  1  1  1  7  7  7  7  7  7  7
  1  1  8  8  8  8  8  8  8  8
  1  9  9  9  9  9  9  9  9  9
 10 10 10 10 10 10 10 10 10 10
```

```
for( r=1; r<=4; r++ )
{
for( c=1; c<=4;c++)
{
if( c<=n-r)p(1);  else  p(r);
}
p("\n");
}
```

$1 <= 2$

n	r	1	r
4	1	3	1
4	2	2	2
4	3	1	3
4	4	0	4

```
1 1 1 1
1 1 2 2
1 3 3 3
4 4 4 4
```

TC

File Edit Run Compile Project Options Debug Break/watch

Line 12 Col 78 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&n);
for(r=1;r<=n;r++)
{
for(c=1;c<=n;c++)
{
if(r==c)printf("%3d",1);  else if(r>c)printf("%3d",2);  else printf("%3d",0);_
}
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC

```
Enter no of rows 3
  1  0  0
  2  1  0
  2  2  1
```

Activate Windows
Go to PC settings to activate Windows.

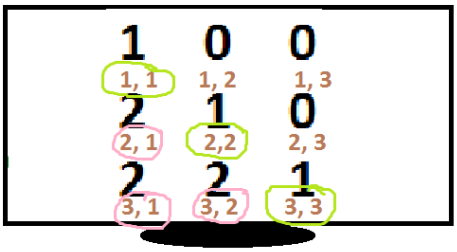
```
Enter no of rows 10
  1  0  0  0  0  0  0  0  0  0
  2  1  0  0  0  0  0  0  0  0
  2  2  1  0  0  0  0  0  0  0
  2  2  2  1  0  0  0  0  0  0
  2  2  2  2  1  0  0  0  0  0
  2  2  2  2  2  1  0  0  0  0
  2  2  2  2  2  2  1  0  0  0
  2  2  2  2  2  2  2  1  0  0
  2  2  2  2  2  2  2  2  1  0
  2  2  2  2  2  2  2  2  2  1
```

Activate Windows
Go to PC settings to activate Windows.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&n);
for(r=1;r<=n;r++)
{
for(c=1;c<=n;c++)
{
if(r==c){textcolor(LIGHTRED);cprintf("%3d",1);}
else if(r>c){textcolor(LIGHTGREEN);cprintf("%3d",2);}
else {textcolor(YELLOW);cprintf("%3d",0);}
}
printf("\n");
}
getch();
}
```

```
Enter no of rows 10
  1  0  0  0  0  0  0  0  0  0
  2  1  0  0  0  0  0  0  0  0
  2  2  1  0  0  0  0  0  0  0
  2  2  2  1  0  0  0  0  0  0
  2  2  2  2  1  0  0  0  0  0
  2  2  2  2  2  1  0  0  0  0
  2  2  2  2  2  2  1  0  0  0
  2  2  2  2  2  2  2  1  0  0
  2  2  2  2  2  2  2  2  1  0
  2  2  2  2  2  2  2  2  2  1
```

```
for( r=1; r<=4; r++ )
{
for( c=1; c<=4;c++)
{
if(r==c)p(1);else if(r>c)p(2);else p(0);
}
p("\n");
}
```



TC

File Edit Run Compile Project Options Debug Break/watch

Line 10 Col 30 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&n);
for(r=1;r<=n;r++)
{
for(c=1;c<=r;c++)printf("*");
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC

```
Enter no of rows 10
*
**
***
****
*****
******
*******
********
*********
**********
```

Activate Windows
Go to PC settings to activate Windows.

```
for( r=1; r<=4; r++ )
{
for( c=1; c<=r;c++)
{
p("*");
}
p("\n");
}
```

r	c=1 to r	
1	1 to 1	*
2	1 to 2	* *
3	1 to 3	* * *
4	1 to 4	* * * *

```
*
* *
* * *
* * * *
```

TC

File Edit Run Compile Project Options Debug Break/watch

Line 10 Col 34 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c,a=1;
clrscr();
printf("Enter no of rows "); scanf("%d",&n);
for(r=1;r<=n;r++)
{
for(c=1;c<=r;c++)printf("%3d",a++);
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC

```
Enter no of rows 10
  1
  2  3
  4  5  6
  7  8  9 10
 11 12 13 14 15
 16 17 18 19 20 21
 22 23 24 25 26 27 28
 29 30 31 32 33 34 35 36
 37 38 39 40 41 42 43 44 45
 46 47 48 49 50 51 52 53 54 55
```

Activate Windows
Go to PC settings to activate Windows.

a=1

```
for( r=1; r<=4; r++ )
{
for( c=1; c<=r;c++)
{
p("  a++  ");
}
p("\n");
}
```

r	c=1 to r	
1	1 to 1	*
2	1 to 2	* *
3	1 to 3	* * *
4	1 to 4	* * * *

Floyd's triangle

```
1
2 3
4 5 6
7 8 9 10
```

TC

File Edit Run Compile Project Options Debug Break/watch

Line 7 Col 55 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c,a;
clrscr();
printf("Enter no of rows "); scanf("%d",&n); a=n*(n+1)/2;
for(r=1;r<=n;r++)
{
for(c=1;c<=r;c++)printf("%3d",a--);
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC

```
Enter no of rows 3
  6
  5  4
  3  2  1
```

Activate Windows
Go to PC settings to activate Windows.

```
Enter no of rows 4
 10
  9  8
  7  6  5
  4  3  2  1
```

Activate Windows
Go to PC settings to activate Windows.

```
Enter no of rows 10
 55
 54 53
 52 51 50
 49 48 47 46
 45 44 43 42 41
 40 39 38 37 36 35
 34 33 32 31 30 29 28
 27 26 25 24 23 22 21 20
 19 18 17 16 15 14 13 12 11
 10  9  8  7  6  5  4  3  2  1
```

Activate Windows
Go to PC settings to activate Windows.

3*4/2=6

n=3

6

5 4

3 2 1

4*5/2=10

n=4

10

9 8

7 6 5

4 3 2 1

TC

File Edit Run Compile Project Options Debug Break/watch

Line 7 Col 46 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c,a=1,b;
clrscr();
printf("Enter no of rows "); scanf("%d",&n);
for(r=1;r<=n;r++)
{
b=a+r-1;
for(c=1;c<=r;c++,a++)if(r%2==0)printf("%3d",b--);else printf("%3d",a);
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC

```
Enter no of rows 4
  1
  3  2
  4  5  6
 10  9  8  7
```

Activate Windows
Go to PC settings to activate Windows.

```
Enter no of rows 10
  1
  3   2
  4   5   6
 10   9   8   7
 11  12  13  14  15
 21  20  19  18  17  16
 22  23  24  25  26  27  28
 36  35  34  33  32  31  30  29
 37  38  39  40  41  42  43  44  45
 55  54  53  52  51  50  49  48  47  46
```

Activate Windows
Go to PC settings to activate Windows.

```
a=1
for( r=1; r<=4; r++ )
{  b=a+r-1;
for( c=1; c<=r;c++,a++ )
{
 if( r%2==0 ) p(b--);
 else  p(a);
}
p("\n");
}
```

r	c	a++
1	12	1 2 3 4 5 6 7
2	123	
3	1234	
4	1	

b--

2+2-1=3

2

7+4-1=10

1
32
456
10987

1
3 2
4 5 6
10 9 8 7

TC

File Edit Run Compile Project Options Debug Break/watch

Line 10 Col 17 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&n);
for(r=n;r>=1;r--)
{
for(c=1;c<=r;c++)printf("%3d",c);
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC

```
Enter no of rows 4
  1  2  3  4
  1  2  3
  1  2
  1
```

Activate Windows
Go to PC settings to activate Windows.

n	= r--	c=1 to r	
4	4	1 to 4	→ 1 2 3 4
	3	1 to 3	→ 1 2 3
	2	1 to 2	→ 1 2
	1	1 to 1	→ 1

TC

Line 18 Col 2 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c,e=2,o=1;
clrscr();
printf("Enter no of rows "); scanf("%d",&n);
for(r=1;r<=n;r++)
{
for(c=1;c<=r;c++)
{
if(r%2==0){printf("%3d",e);e+=2;}
else  {printf("%3d",o);  o+=2;}
}
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC

```
Enter no of rows 4
  1
  2  4
  3  5  7
  6  8 10 12
```

Activate Windows
Go to PC settings to activate Windows.

Page: 23 of 23 Words: 0 130%

$\frac{e+2}{2}$ $\frac{o+2}{1}$

1
2 4
3 5 7
6 8 10 12

```
Enter no of rows 10
  1
  2   4
  3   5   7
  6   8  10  12
  9  11  13  15  17
 14  16  18  20  22  24
 19  21  23  25  27  29  31
 26  28  30  32  34  36  38  40
 33  35  37  39  41  43  45  47  49
 42  44  46  48  50  52  54  56  58  60
```

```
*
**
***
****
*****

*
* *
* * *
* * * *

* * * *
* * *
* *
*

*
* *
* * *
* * * *
* * * *
* * *
* *
*

*
* *
*   *
*     *
*       *
*     *
*   *
* *
*
```